MRUNAL HOLE

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EDUCATION

The Ohio State University Columbus, OH
Master of Science, Computer Science and Engineering; CGPA: 3.82/4 August 2024 - May 2026
Courses: Speech and Language Processing, Artificial Intelligence, Data Mining

Savitribai Phule Pune University Pune, India
Bachelor of Engineering in Information Technology; CGPA: 9.30/10 August 2018 - June 2022
Courses: Operating Systems, Theory of Computation, Database Management, Object-Oriented Programming

TECHNICAL SKILLS

Programming Languages: C, C++, Python, R, Java, Data Structures, SQL, JavaScript, HTML5, CSS3

Technologies: Databricks, Snowflake, Apache Spark, AWS, ADF, Azure Synapse Analytics, Tableau, Power BI, Airflow, Azure Data Studio, Microsoft SQL Server, Azure DevOps, Git, Cursor, Django, MongoDB, Hadoop, Node.js, NoSQL, RDBMS, PostgreSQL, RESTful API, Docker, Spring Boot

ML Frameworks: Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, OpenCV, Seaborn, Scikit-learn, Hugging Face, Langchain

WORK EXPERIENCE

ParaWave Marysville, OH
Full Stack Intern May 2025 – August 2025

- Built and optimized a web app with custom Dart actions and widgets, integrating **FastAPI** and Node.js as backend systems. Leveraged **AWS Lambda**, RDS, S3, and Glue to automate video streaming workflows using CloudWatch and containerized microservices like Docker and Kubernetes, and Kafka, including **pre-signed URLs**, dynamic UI rendering, and **RESTful API** integration.
- Fine-tuned AI models for object detection and segmentation (**3 sec avg detection**), evaluated pipelines, and model training workflows to enhance computer vision accuracy by **29%**, while also applying cybersecurity principles.
- Integrated server with RTMP, RTSP, WebRTC, and HLS protocols for real-time video streaming, enabling low-latency (**1-2 sec**) playback, multi-stream handling, and seamless client access at scale.

Persistent Systems Ltd Pune, India
Senior Software Engineer April 2024 – July 2024

- Engineered and managed notebooks, views, tables in **Databricks** (Spark SQL, Python), and procedures in Azure Synapse Analytics.
- Implemented data load strategies including Common Table Expressions (**CTEs**) and temporary tables in SQL, managing both full and incremental data loading processes, and reduced data redundancy by **50%**.
- Streamlined data workflows via Azure Data Factory (ADF), uploading files to the Data Lake and extracting and transforming data for Master Data Management (**MDM**) systems via API, reducing transfer time by **40%**.
- Developed and maintained unit testing documentation to validate data integrity and ensure 100% accuracy in data processing workflows, which contributed to a **30%** decrease in processing errors.

Software Engineer July 2022 – April 2024

- Improved enterprise data warehouse by modifying views, tables, and stored procedures, boosting query performance by **15%**.
- Implemented Continuous Integration/Continuous Deployment (**CI/CD**) with ADF using JSON for **20+** pipelines, maintained through Agile methodology and Jira sprint planning, reducing deployment time by **30%** and increasing frequency by **25%**.
- Ensured 100% database consistency across environments via comparison and sync using Azure Data Studio, SSMS, and DevOps.

Software Engineer Intern February 2022 – July 2022

- Translated SQL server **Stored Procedures** into **DBT models** on MySQL and PostgreSQL, ensuring seamless migration.
- Facilitated the migration of over **30 Tableau** reports from SQL Server to **Snowflake**, creating new data sources and building reports from scratch resulting in a **30%** improvement in reporting efficiency for 50 stakeholders.

PROJECTS

Multimodal AI Video Generation

- Analyzed multimodal architecture designs integrating CLIP, 3D VAEs, Diffusion models, Vision Transformers, and transformer-based pipelines, with **Retrieval-Augmented Generation** (RAG) supported by vector databases such as **ChromaDB** for context-aware text-to-video synthesis.
- Evaluated **U-Net**/Transformer architectures for optimizing content delivery in behavioral intervention applications.
- Investigating temporal alignment techniques and structure conditioning methods to ensure consistent gesture representation and natural movements in AI-generated content and content-aware motion adapters for creating personalized behavioral health videos for children.

DigitalAgEdu - AI Curriculum Generation Toolkit

Python, Typer, Pydantic, Jinja2, YAML

- Built a config-driven Python CLI system to auto-generate structured AI curricula using schema-validated inputs and template-based rendering.
- Designed modular data models and parsers for **YAML/JSON** configs, enabling reproducible, extensible content generation pipelines.
- Implemented end-to-end automation from validated configuration → rendered artifacts, packaged for pip-based deployment.
- Architected the system for scalability, testability, and **CI/CD** readiness, following production-grade software practices.

CERTIFICATIONS

AZ-900: Microsoft Azure Fundamentals - Link SnowPro Core COF - CO2 certification - Link

SELECTED PUBLICATION

May 2022 – Vehicle Tracking System with Anti-theft Security and Parental Control (paper link)